Sustainable Mobility

Module Overview

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2025/2026 – 2nd semester

A BRIEF PRESENTATION OF MYSELF…

Coordination & Teaching: Margarida C. Coelho

- Full Professor at the Dept. Environment & Planning of U. Aveiro
- Field of expertise: Transportation, Energy and Environment
- Course Director of the Master degree on Smart Mobility
- Local coordination of the research on transport & mobility at ECIU
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FACT

Transport & mobility:

One of the most relevant sectors of society, as it interferes in our daily lives (we have and will always need to move people and goods).

FACT

The mobility area is undergoing very interesting disruptive transformations in recent times (such as: electric, autonomous and shared mobility; mobility-as-a-service; vertical mobility)

It is one of the most relevant sectors of society, as it interferes in our daily lives (we have and will always need to move people and goods).

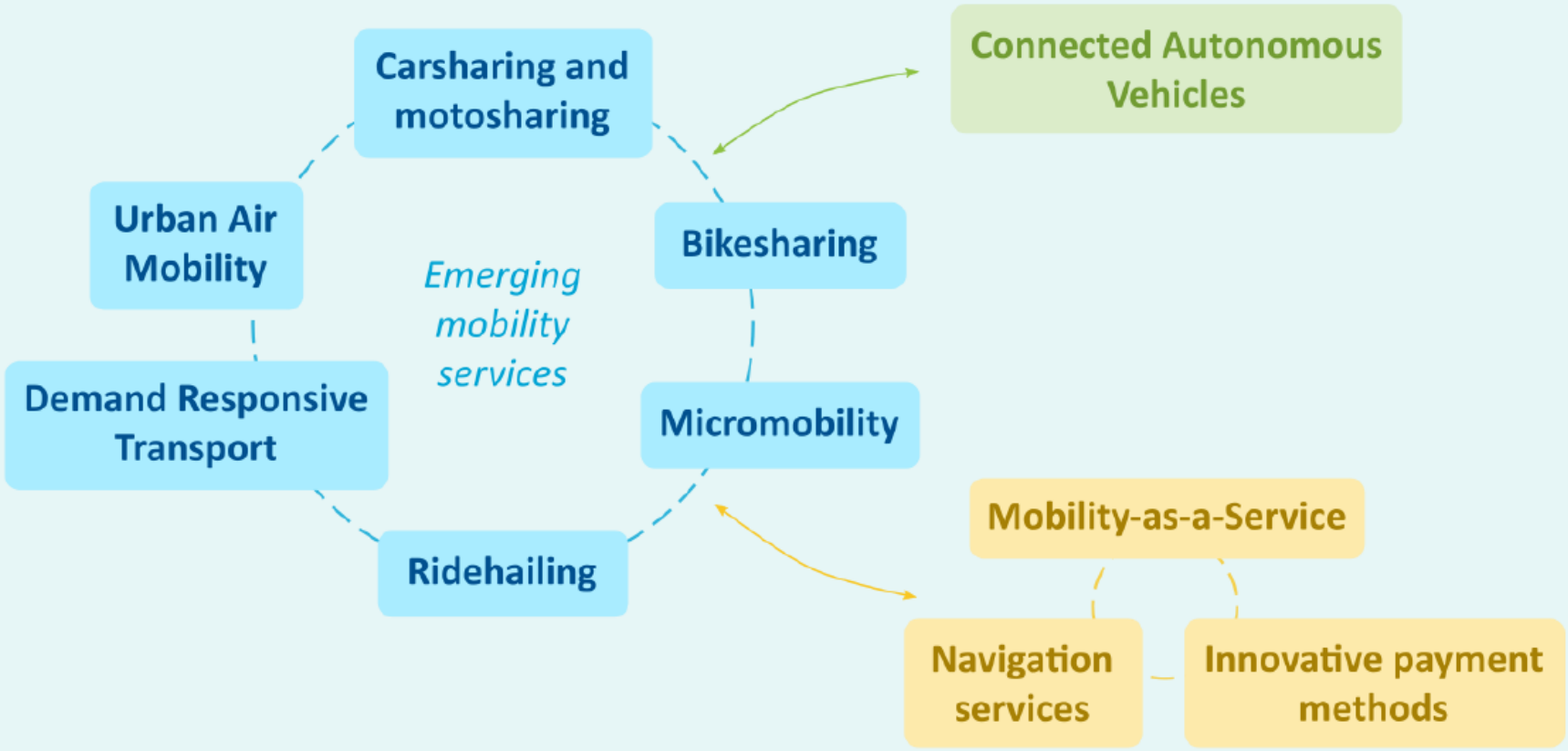
We need professionals from different engineering backgrounds.

Motivation

Ref: TRB (Your future in transportation, 2020)

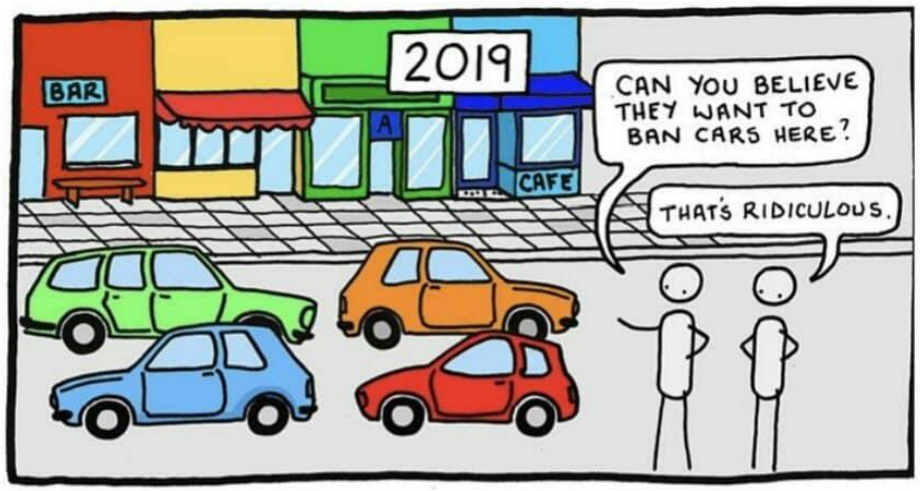
Motivation

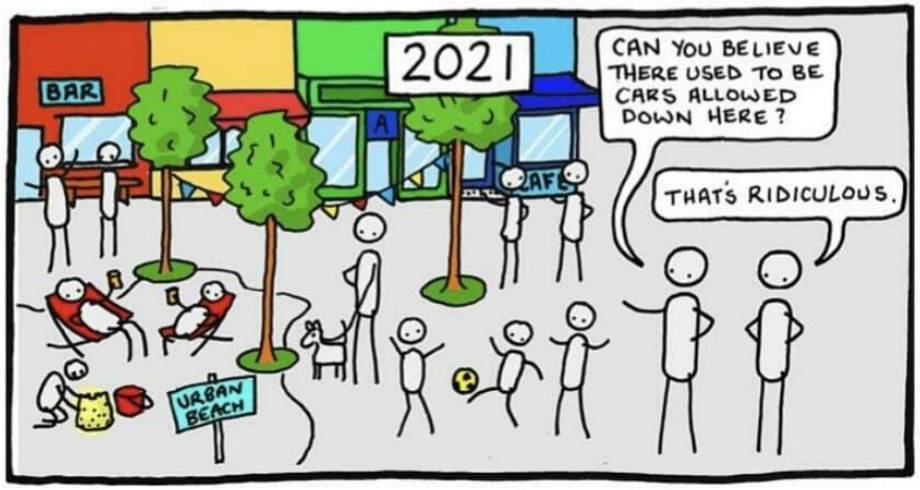
Supply innovations trends

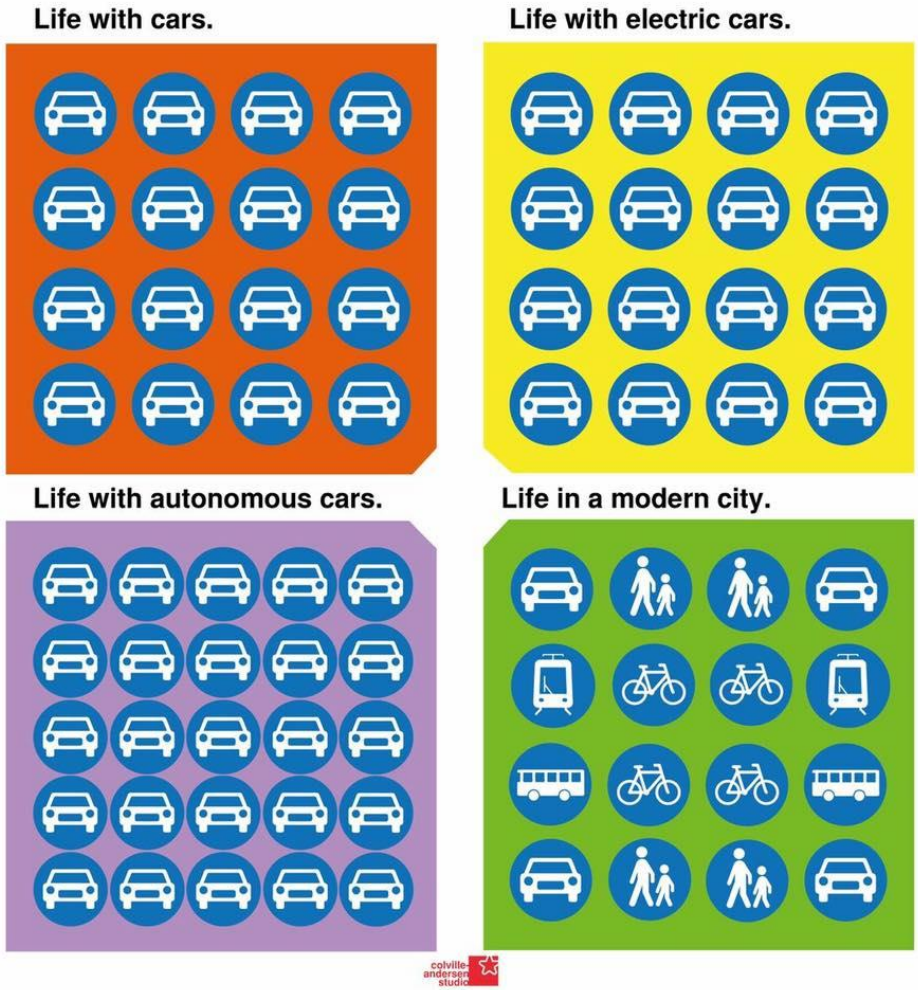


Fonte: MOMENTUM (New Mobility Options and Urban Mobility: Challenges and Opportunities for Transport Planning and Modelling, 2019)

What is the mobility that we really want for our future?







Ref : Colville-Andersen studio

Why this "Sustainable Mobility" module is useful?

Future possibilities: fleet managers; municipalities; transport companies; ICT companies; energy agencies; energy consulting companies; …

Relevant module for those who want to follow the MSc. Degree on Smart Mobility

OBJECTIVES

It is intended that the students should be able to:

-Understand the current state of the transportation sector from the point of view of energy and environment;

-Analyze urban mobility problems and identify solutions for sustainable mobility in terms of energy and environment;

-Identify alternative fuels and alternative propulsion technologies.

CONTENTS

- Present situation of mobility: Europe and Portugal

- Challenges of the transportation sector

- Impacts:
  - Traffic congestion;
  - Safety;
  - Greenhouse gases emissions (Climate change) and pollutant emissions;
  - Noise

- Sustainable Mobility Measures

- Alternative fuels and propulsion modes

GROUP 1 - STRUCTURE

- **Timetable:**

  Monday, 10h30-12h

  Thursday, 17h30-19h

Grupo	Período	2ª f	5ª f	Dias	
1	1.º (9 fevereiro a 12 março)	Apresentação; Introdução - Estado do setor dos transportes	Combustíveis & Modos alternativos de propulsão	09/fev	12/fev
			Desafios do setor dos transportes	**2ª f de Carnaval**	19/fev
		Emissões GEE e poluentes (COPERT)	Emissões GEE e poluentes (COPERT)	23/fev	26/fev
		Medidas de mobilidade sustentável	Medidas de mobilidade sustentável	02/mar	05/mar
		Combustíveis & Modos alternativos de propulsão	Avaliação (Teste)	09/mar	12/mar

- **Evaluation: Final Test (March 12)**

Multiple choice questions + COPERT exercise

COPERT Download at: https://copert.emisia.com/

GROUP 2 - STRUCTURE

- **Timetable:**

  Monday, 10h30-12h

  Thursday, 17h30-19h

2.º (16 março a 23 de abril)	Apresentação; Introdução - Estado do setor dos transportes	Desafios do setor dos transportes	16/mar	19/mar	
	Desafios do setor dos transportes	Emissões GEE e poluentes (COPERT)	23/mar	26/mar	
	Emissões GEE e poluentes (COPERT)		30/mar	**Férias Páscoa**	
	Medidas de mobilidade sustentável	Medidas de mobilidade sustentável	13/abr	16/abr	
	Combustíveis & Modos alternativos de propulsão	Avaliação (Teste)	20/abr	23/abr	

- **Evaluation: Final Test (April 23)**

Multiple choice questions + COPERT exercise

COPERT Download at: https://copert.emisia.com/

GROUP 3 - STRUCTURE

- **Timetable:**

Monday, 10h30-12h
Thursday, 17h30-19h

3.º (04 maio a 01 junho)	**Apresentação; Introdução - Estado do setor dos transportes**	**Desafios do setor dos transportes**	04/mai	07/mai
	Emissões GEE e poluentes (COPERT)	**Emissões GEE e poluentes (COPERT)**	11/mai	14/mai
	Medidas de mobilidade sustentável	**Ruído - Noise Capture Party**	18/mai	21/mai
	Medidas de mobilidade sustentável	**Combustíveis & Modos alternativos de propulsão**	25/mai	28/mai
	Avaliação (Teste)		01/jun	

- **Evaluation: Final Test (June 1)**

Multiple choice questions + COPERT exercise
COPERT Download at: https://copert.emisia.com/

WHAT I EXPECT…

- Active participation
- Participant-centered learning:
  - Learn from you
  - Learn from peers
- Engage you and have fun
- Rigour & Academic honesty
- Your feedback

REFERENCES

- Slides

- EEA (2024). TERM report, European Environment Agency, https://www.eea.europa.eu/en/analysis/publications/sustainability-of-europes-mobility-systems

- EEA (2025) Europe's environment 2025 - Europe's environment and climate: knowledge for resilience, prosperity and sustainability, https://www.eea.europa.eu/en/europe-environment-2025/main-report

- EC (2025) EU Transport in Figures – Statistical Pocketbook 2025. European Commission.